

Dhiraj Kumar

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PROFESSIONAL SUMMARY

Full-stack and AI engineer who designs and ships production-grade, AI-integrated systems end-to-end — from multi-agent LLM pipelines and encrypted, read-only database-access layers to automated content-processing platforms. Comfortable owning the full stack: system architecture, authentication, background-job orchestration, caching, and real-time streaming. Self-taught from first principles; independently shipped 3 production applications in 6 months. Targeting Software Engineer, Full-Stack, Backend, or AI Engineer roles.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript

Frontend: React, Next.js, Redux, Zustand, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, Authentication & Authorization (OAuth)

Databases: PostgreSQL, MySQL, MongoDB, Drizzle ORM

AI/LLM Engineering: OpenAI, Groq, Google Gemini, Vercel AI SDK, TF-IDF Schema Filtering, Multi-Agent Pipelines, Tool Calling, Prompt Engineering

Infra & Deployment: Inngest (event-driven background jobs), Vercel

Tools: Git, GitHub, Postman, Chrome Extension APIs (Manifest V3), Web Scraping (RSS, Cheerio, Readability.js)

FREELANCE EXPERIENCE

Freelance Full-Stack Developer — Remote

2025 – Present

- Built and shipped a production marketing website for Annu Placement, a live client, using Next.js, Tailwind CSS v4, and animation libraries, taken from requirements through deployment.
- Designed and built a custom lead-capture pipeline from scratch that receives inbound leads in real time and writes them directly to Google Sheets and Telegram, giving the client instant visibility with zero manual data entry.
- Delivered a system actively used by a real customer in production, handling live lead traffic reliably end-to-end.

PROJECTS

QueryMate — AI-Powered Database Query Platform

Mar 2026 – Jun 2026

Next.js, TypeScript, PostgreSQL, Drizzle ORM, Vercel AI SDK, Groq

- Engineered a production-safe natural-language-to-SQL platform via a 3-agent LLM pipeline (Classification → Tool-Call → Answer) with OAuth authentication (BetterAuth) and sliding-window conversation memory, letting non-technical users query PostgreSQL, MySQL, and MongoDB in plain English.
- Enforced strict read-only access per database engine and encrypted connection strings with AES-256-GCM bound to user identity, closing unauthorized write and data-leak paths.
- Reduced LLM context bloat through TF-IDF schema filtering with foreign-key expansion, keeping large multi-table schemas within token limits without losing query accuracy.
- Hand-built the first version entirely through manual AI orchestration — custom agent routing, tool-calling logic, and manually handled SSE streaming — then migrated to the Vercel AI SDK for production stability, retaining the manual prompt-engineering and streaming pipeline as engineering differentiators.

BlogDrop — AI Engineering Blog Aggregator

Jun 2026 – Present

Next.js, TypeScript, PostgreSQL, Drizzle ORM, Inngest, Groq/OpenAI API

- Architected an AI-powered blog-aggregation platform that autonomously discovers, harvests, and deduplicates engineering content from RSS feeds and sitemaps across dozens of sources.
- Built a fault-tolerant, event-driven ingestion pipeline with Inngest, achieving a 96%+ successful-harvest rate across a 500-article batch — the only 21 failures traced to Inngest's execution/rate-limit ceiling, not pipeline defects.
- Cut LLM cost and rate-limit failures by benchmarking OpenAI, Groq, and Gemini and engineering batch processing with retry logic, using AI summarization to auto-categorize content.

SmartForm — AI-Powered Chrome Extension for Intelligent Form Auto-Filling

Feb 2026 – Mar 2026

TypeScript, WXT Framework, Tailwind CSS, Gemini AI API, Chrome Extension APIs

- Built a Chrome extension (Manifest V3) that autonomously detects and fills web forms using context-aware AI-generated data, removing repetitive manual entry on dynamic single-page apps.
- Engineered prompt and JSON-schema constraints for the Gemini API to produce deterministic output, with defensive parsing and fallback handling for malformed LLM responses.
- Designed a privacy-first, local-storage-only architecture, ensuring zero external transmission of user-entered form data.

ACHIEVEMENTS & OPEN SOURCE

- Independently designed, built, and shipped 3 production AI-integrated applications in 6 months alongside full-time academics.
- Engineered BlogDrop's harvesting pipeline to a 96%+ success rate at 500-article scale, with failures isolated entirely to platform rate limits, not application logic.
- Maintains public GitHub repositories for all projects with documented architecture decisions; build-in-public technical content reaching 300+ post impressions.

EDUCATION

SMD College, Punpun, Patna

Mar 2023 – Mar 2025

Intermediate (PCM), Board Examination